

PXE IMAGING A WORKSTATION

Six stages take this machine from powered off to imaged, named from the desk label, and joined to **ITS.lan**. Work top to bottom. Do not skip a stage, and do not change anything in BIOS Setup that this guide does not tell you to change.

01 LINK UP Cable in, light on	02 BOOT MENU Tap the key at the logo	03 PXE MENU IPv4 network → Windows	04 1 TB DISK Pick Disk 1, then image	05 NAME Copy the desk label	06 JOIN ITS.LAN Domain join, reboot
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BEFORE YOU TOUCH THE POWER BUTTON

- ☐ Ethernet cable is seated in the wall port **and** the workstation's network port. **PXE does not work over Wi-Fi.**
- ☐ The link light next to the network port comes on within about 5 seconds of power. No light means no PXE — swap the cable first.
- ☐ Find the label taped to the desk in front of this workstation. It reads something like **CIT6 R1C3**. That is this machine's name.
- ☐ These workstations have **two drives**. Windows goes on the **1 TB hard drive**, never the small SSD.
- ☐ Nothing on this machine needs to be saved. **Imaging erases the drive you select.**

COPY THE DESK LABEL HERE BEFORE YOU START

EXAMPLE CIT6 R1C3

RULES OF THE ROOM

1. **One machine, one student.** Image only the workstation you were assigned. Wiping someone else's machine means their work is gone.
2. **Never cut power during Stage 04.** A half-written disk means starting the whole thing over.
3. **Stuck more than five minutes? Raise your hand.** Guessing in BIOS costs the whole class time.
4. **Read errors out loud, word for word.** A PXE code tells your instructor exactly what broke — "it didn't work" doesn't.

BOOT MENU KEY _____	PXE MENU ENTRY _____	WINDOWS TASK SEQUENCE _____	DOMAIN ITS.lan
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INSTRUCTOR FILLS THE FIRST THREE IN BEFORE CLASS

DONE LOOKS LIKE THIS

- | | | | |
|--|---|---|---|
| 01
Windows boots from the local disk — no PXE screen | 02
hostname prints the label taped to the desk | 03
Full device name ends in .ITS.lan | 04
Drive C: sits on the 932 GB disk, not the SSD |
|--|---|---|---|

01

LINK UP

BRING THE LINK UP

GOAL – A LIT NETWORK PORT BEFORE ANYTHING ELSE HAPPENS

1. Push the ethernet cable into the workstation's network port until it clicks. Do the same at the wall port.
2. Press the power button once, then look at the small light beside the network port — it should come on within about five seconds.

NO LIGHT Try a different cable. Then a different wall port. Do not continue to Stage 02 without a link light — PXE will fail every time.

02

BOOT MENU

OPEN THE ONE-TIME BOOT MENU

GOAL – INTERRUPT THE NORMAL STARTUP AND PICK WHERE IT BOOTS FROM

As soon as the manufacturer logo appears, **tap** the boot menu key about once per second until the menu opens. Tapping works better than holding the key down.

DELL PRECISION

F12

HP

F9

LENOVO

F12

ACER / ASUS

F12 / ESC

MISSED IT If Windows starts loading instead, you were too slow. Hold the power button 5 seconds, power back on, and try again. Almost everyone misses it the first time.

DO NOT Do not enter BIOS Setup (usually **F2** or **Del**) and start changing settings. If there is no network option in the menu, that is a Stage 02 fix on page 5 — raise your hand.

03

PXE MENU

BOOT FROM THE NETWORK, THEN CHOOSE WINDOWS

GOAL – REACH THE DEPLOYMENT MENU SERVED OVER THE NETWORK

Use the arrow keys to highlight the **IPv4 network** entry and press **Enter**. Ignore any IPv6 entry — it will not work here.

Boot Menu

Windows Boot Manager

UEFI: Hard Drive

> UEFI: IPv4 Network / PXE – Onboard NIC

UEFI: IPv6 Network / PXE – Onboard NIC

USB Storage Device

Enter Setup

WORDING VARIES BY MANUFACTURER – LOOK FOR NETWORK, PXE, OR NIC TOGETHER WITH IPV4

Next comes plain text on black: the machine asking the network for an address and downloading the boot file. **Let it run.**

>>Start PXE over IPv4.

Station IP address is 10. . . . <- got an address, good

Downloading NBP file...

WAIT, THEN GO This can take up to a minute — do not press keys or reboot. When the deployment menu loads, pick the **Windows** entry named at the bottom of page 1. If a **PXE-E** code appears instead, write it down and use the table on page 5.

04

1 TB DISK

INSTALL TO THE 1 TB HARD DRIVE

GOAL – WINDOWS LANDS ON THE 932 GB DISK, NOT THE 238 GB SSD

These workstations have two drives, and the installer will offer you both. **You want the big one.** The size is how you tell them apart — the number you want starts with a **9**.

Where do you want to install Windows?

```

Disk 0  238.0 GB  Micron 1100 SATA 256GB  <- SSD. Leave it alone.
> Disk 1  932.0 GB  ST1000DM003-1ER162  <- Install here.
  
```

DISK NUMBERS CAN MOVE BETWEEN MACHINES – GO BY THE **SIZE**, NOT THE DISK NUMBER**POINT OF NO RETURN**

The next prompt erases the disk you selected. Before you confirm: the desk label matches your assignment, **and** the highlighted disk says **932** GB. Installing to the SSD means the whole build is redone.

THEN WAIT

The machine restarts itself several times on its own — that is normal. Do not power off, do not unplug the ethernet cable, do not press keys to "speed it up," do not walk away.

05

NAME

NAME IT FROM THE DESK LABEL

GOAL – HOSTNAME MATCHES THE TAPED LABEL, CHARACTER FOR CHARACTER

TAPED TO THE DESK
CIT6 R1C3

→
TYPE IT

Computer name

CIT6-R1C3**THE ONE CHANGE**

The label has a **space**. A computer name cannot. Type a **hyphen** where the space is: **CIT6 R1C3** on the desk becomes **CIT6-R1C3** on the machine. Nothing else changes.

WHAT THE LABEL MEANS

CIT6

The room. Same on every machine in here.

R1

Row 1 — rows count front of room to back.

C3

Column 3 — seats count left to right.

THREE WAYS STUDENTS GET THIS WRONG

CIT6-R3C1

Row and column swapped. **R** is always the row, **C** is always the column. Read left to right.

CIT6-RIC3

Letter I instead of the digit 1. Look at the label again before you press Enter.

CLT6-R1C3

Lowercase L instead of capital I. It is **C-I-T-6**, for the room.

RULES

15 characters maximum · no spaces · letters, digits and hyphens only. The wizard usually asks during setup; if it never asked, use **Settings → System → About → Rename this PC**, then restart.

06

JOIN ITS.LAN

JOIN THE ITS.LAN DOMAIN

GOAL – THE MACHINE IS A MEMBER OF ITS.LAN AND SHOWS UP IN ACTIVE DIRECTORY

1. Sign in with the local account the image created.
2. Open **Settings** → **System** → **About**, then click **Domain or workgroup**.
3. On the **Computer Name** tab, click **Change...**
4. Check that **Computer name** already reads the desk label with a hyphen. Fix it here if it doesn't.
5. Under **Member of**, select **Domain** and type it in full: `ITS.lan`
6. Enter the join username and password your instructor reads out. **Do not write them on this sheet.**
7. You should see a welcome message. Click **OK**, then **Restart now**.
8. After the reboot, sign in as `ITS\yourusername`.

Computer Name/Domain Changes

Computer name: `CIT6-R1C3`
Full computer name: `CIT6-R1C3.ITS.lan` <- this is what you want to see
Member of: (o) Domain: `ITS.lan` () Workgroup: WORKGROUP

IF THE JOIN FAILS, DO NOT RETRY OVER AND OVER – COPY THE MESSAGE DOWN AND USE THE TABLE ON PAGE 5

VERIFY — OPEN COMMAND PROMPT (PRESS `WIN`, TYPE `CMD`, PRESS ENTER)

CHECK 1 – THE NAME

`hostname`

Prints the desk label with a hyphen, e.g. `CIT6-R1C3`.

CHECK 2 – THE DOMAIN

`systeminfo | find "Domain"`

Shows `ITS.lan`, not `WORKGROUP`. Takes a few seconds to return.

CHECK 3 – THE DISK

`diskmgmt.msc`

Drive **C:** sits on the **932 GB** disk. The 238 GB SSD is a separate, untouched disk.

CHECK 4 – REACH THE DC

`ping <dc name>`

Replies with no timeouts. Ask your instructor for the DC name.

Last check is your instructor's: the computer object appears in **Active Directory Users and Computers** under the correct OU.

STATION LOG — FILL IN A ROW AS YOU FINISH EACH MACHINE

DESK LABEL	COMPUTER NAME	IMAGED BY	ON 932 GB	ITS.LAN	INSTRUCTOR

WHEN SOMETHING GOES WRONG

WRITE THE MESSAGE DOWN BEFORE YOU ASK

WHAT YOU SEE	WHAT IT MEANS	WHAT YOU DO
No network / PXE entry in the boot menu	Network boot is turned off in the machine's firmware.	Raise your hand. An instructor enables the network stack. Do not change UEFI settings yourself.
PXE-E51 No DHCP or proxyDHCP offers received	The machine never got an IP address. Cable, port, or wrong VLAN.	Check the link light → swap the cable → try a known-good wall port → then tell your instructor.
PXE-E53 No boot filename received	It got an address, but nothing answered with a boot file.	Report it. Usually the wrong port/VLAN or the deployment server needs attention.
PXE-E61 Media test failure, check cable	The port sees no physical link at all.	Reseat both ends of the cable, then swap the cable for a known-good one.
"Start PXE over IPv4" flashes, then Windows loads	PXE timed out and the machine fell through to the hard drive.	Reboot, get into the boot menu faster, and select the IPv4 network entry again.
Only one disk shows in the installer	A drive is unplugged, failed, or not seen by the controller.	Stop. Do not install to whatever is showing. Tell your instructor which size you can see.
You installed to the 238 GB SSD	Wrong disk. C: is on the SSD instead of the 1 TB drive.	Tell your instructor, then re-image from Stage 03 and pick the 932 GB disk this time.
Imaging stops partway or throws an error	Failed download, or a problem with the disk.	Write the error text down word for word, then get an instructor. Do not retry blindly.
Name rejected: invalid characters	You typed the space from the desk label.	Replace the space with a hyphen: CIT6-R1C3 . No other characters change.
Join fails: the domain ITS.lan could not be contacted	The machine isn't using the domain controller for DNS.	Run <code>ipconfig /all</code> . The DNS server listed should be the domain controller. If it isn't, report it.
Join fails: an account with that name already exists	An old computer object with this name is still in Active Directory.	Tell your instructor — they remove the stale object, then you retry the join.
You joined with the wrong name	Fixable, but it must be reported.	Settings → System → About → Rename this PC , restart, and have an instructor confirm AD updated.

QUICK REFERENCE — THE THREE THINGS THAT MUST BE RIGHT

NAME — FROM THE DESK LABEL

CIT6-R1C3

Room, row, column. Space on the label becomes a hyphen on the machine.

DOMAIN — TYPED IN FULL

ITS.lan

Full name becomes CIT6-R1C3.ITS.lan. Sign in as ITS\yourusername.

DISK — THE BIG ONE

932 GB

The 1 TB Seagate. Never the 238 GB Micron SSD.